

Unit 7 Test Study Guide

(Polynomials & Factoring)

Name: _____

Date: _____ Per: _____

Topic 1: Writing Polynomials in Standard Form

Directions: Write each polynomial in standard form.

1. $a^2 + 3a^4 - 7a + 2a^3 - 4$

2. $8y^3 - 3xy^2 - x^2y + 2x^3$

Topic 2: Classifying Polynomials

Directions: Classify each polynomial by degree and number of terms.

3. $3x + 12$

4. 24

5. $-7x^2 + 4x + 1$

6. $3x^4 - x^3 + 5x^2 + x - 7$

Topic 3: Simplifying Polynomials

Directions: Simplify each polynomial. Write all answers in standard form.

7. $(2x^2 + 3x + 2) - (x^2 - 4x - 1)$

8. $2a^3(5a^2 - 7a) + (3a^4 - a^5)$

9. $5c^2d(9c^2d^2 - 4c^3d - 2)$

10. $-4y(5y - 11) - 2y(3y + 10)$

11. $(x - 1)(4x - 3)$

12. $(5y - 1)^2$

13. $(2x + 5y)(x - 3y)$

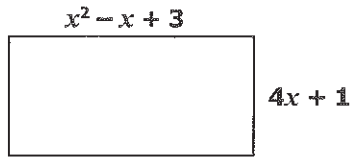
14. $(x + 4)(4x - 2) - 2x^2$

15. $\frac{-8x^5 + 28x^4 - 4x^3}{4x^3}$

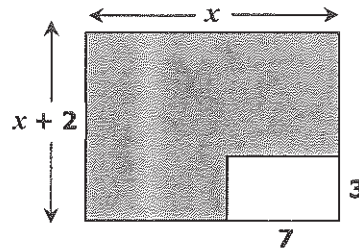
16. $\frac{27c^4d^7 + 15c^3d^5}{3c^2d^3}$

Topic 4: Geometric Applications

- 17.** Write an expression to represent the area of the rectangle below in simplest form.



- 18.** Write an expression to represent the area of the shaded region in simplest form.



Topic 5: Factoring Polynomials

Directions: Factor each polynomial completely.

19. $54x^3y - 18x^2y^2 + 27xy^2$

20. $9x^2 - 49y^2$

21. $64y^4 - 16y^6$

22. $n^3 - 5n^2 - 50n$

23. $4x^3 + 12x^2 - 25x - 75$

24. $3k^2 - 57k + 180$

25. $4m^2 - m - 14$

26. $8r^2 + 16r - 10$

- 27.** If the area of a rectangle can be represented by $(x^2 - 5x - 14)$, what two expressions could represent the dimensions of the rectangle?

Topic 6: Dividing a Polynomial by a Binomial

Directions: Find each quotient.

28. $(x^2 - 14x + 48) \div (x - 6)$

29. $(p^2 - 9) \div (p - 3)$

30. $\frac{a^2b + a^2c - 5b - 5c}{b + c}$